

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CE) – Elective I

March-April, 2018

CE318: Software Project Management & Quality Assurance (SPM & QA)

Date: 05.04.2018, Thursday

Time: 1:30 p.m. To 4:30 p.m.

Maximum Marks: 70

Instructions:

1. Section I and II must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Figures to the right indicate marks.

SECTION-I

Q-1 Answer the following questions.

- (a) Define Term Project. How Software Project is differing from other non-IT Project? [03]
- (b) What is significance of Software Project team structure? Enlist types of team structure. [02]
- (c) Why Software Project fails? [02]

Q – 2 (a) Why Risk Management is important in Software Project? Which are types of Risk involved in Software Project? [04]

Q – 2 (b) Justify why Resources are required in Project Management ? Which are types of resources you can add in MS Project Resources sheet? [05]

OR

Q – 2 (b) In your opinion, do you think that, software project managers update their knowledge with respect to latest technologies? What are your immediate tasks as the Software Project Manager when you are appointed? [05]

Q – 2 (c) Consider a software project with 5 tasks T1-T5. Duration of the 5 tasks (in days) is 15,10,12,25 and 10 respectively. T2 and T4 can start when T1 is complete T3 can start when T2 is complete. T5 can start when both T3 and T4 are complete. [05]

i) Draw the Gantt Chart and PERT charts for the project

ii) When is the latest start date of the task T3? What is the slack time of the task T4? What is the slack time of the task T2?

ii) How Gantt Chart is different from PERT chart?

Q – 3 (a) What is relationship between Time, Scope and Cost of Project? Explain the reasons behind the following assertion "Adding more manpower to a late project makes it later". [05]

Q – 3 (b) What does Halstead's length and volume metric represent conceptually? How according to Halstead is the effort dependent on program volume? [04]

OR

Q – 3 (b) Discuss the aspects, which need to be considered at the pre-feasibility and feasibility study of a project, What factors and issues need to be considered in financial, economic and technical evaluation? [04]

- Q – 3 (c)** How to do tracking in MS Project? Explain significance of Baseline in MS Project [05]
planning and tracking phase. How many number of Baseline you can create in MS Project? Why it's not advisable to reset Baseline more frequently?

SECTION – II

Q - 4 Answer the questions below.

- (a) Justify the statement whether it is true or false “Debugging is not testing, but [02]
always occurs as a consequence of testing”.
- (b) How to do you define word Quality for Software? What is difference between [02]
Software Quality Control and Software Quality Assurance?
- (c) What is difference between reengineering and reverse engineering? How they are [03]
related with software maintenance and change management?
- Q – 5 (a)** What are the types of white box testing techniques? Design path coverage and [05]
statement coverage based test suite for the following Euclid's GCD Program
- ```
int computeGCD(x,y)
int x,y;{
while (x!=y){
if(x>y)then
x=x-y;
else y=y-x;}
```
- Q – 5 (b)** Which are various techniques to do Black box testing? Explain all in brief. Design [05]  
black box test suites for a function that checks whether a character string (of up to  
25 characters in length) is a palindrome.
- Q – 5 (c)** What is change control and version control? How it is related with Software [04]  
Configuration Management?

**OR**

- Q – 5 (a)** What is Cyclomatic complexity? Determine the Cyclomatic Complexity of the sort [05]  
function below.
- ```
void sort(int a[],int n)
{ int i , j;
for(i=0;i<n-1;i++)
for(j=i+1;j<n;j++)
if(a[i]>a[j])
{ Temp=a[i];
a[i]=a[j];
a[j]=temp;} }
```
- Q – 5 (b)** Explain difference between Unit and System Testing? Why Unit testing is club [05]
with Coding? How Alpha testing is differing from Beta and Acceptance testing?
- Q – 5 (c)** What are the different types of maintenance that a software product might need? [04]
Why is this maintenance required?

- Q – 6 (a)** What do you understand by Key Process Areas (KPA), in the context of SEI CMM? Would an Organization encounter any problem, if it tries to implement higher level SEI CMM KPAs before achieving lower level KPAs? **[05]**
- Q – 6 (b)** Distinguish Software Verification and Software Validation? When during software lifecycle are verification and validation performed? Can one be used in other place? **[05]**
- Q – 6 (c)** Define Alpha, Beta and Acceptance Testing. How Alpha and Beta Testing is differing from Acceptance Testing? **[04]**

OR

- Q – 6 (a)** Why it is important for a Software Development Organization to obtain ISO 9001 certification? With the help of examples discuss the types of Software Organization to which ISO9001,ISO9002,ISO9003 standards are respectively applicable **[05]**
- Q – 6 (b)** Why Software Quality Assurance is called as an umbrella activity? **[04]**
- Q – 6 (c)** What is difference between Reliability and Quality? Justify the statement “highly reliable software can be termed as a good quality software”. **[05]**